

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	P. Sanjeev	Reg. No.:	192324231
Section / Batch:	AI & DS E 2023	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I, P. Sanjeev (192324231) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Sanjeev

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Linear search - Target immediately After First Elements.

Given,

dataset : [60, 25, 33, 41, 52, 77, 89]

Target element :- 25

Algorithm:-

Step 1: Start from first element.

Step 2: compare current element with target.

Step 3: If equal, stop & return its index.

Step 4: otherwise move to next element.

Step 5: Repeat until target is found on list ends.

→ Array : [60, 25, 33, 41, 52, 77, 89]
 ?

$A[i] == 25$? No, $i++$

→ [60, 25, 33, 41, 52, 77, 89]
 ?

$A[i] == 25$? Yes, return index.

Target found, search stops.

Number of comparisons.

comparison 1 $\Rightarrow 60 == 25$

comparison 2 $\Rightarrow 25 == 25$

Total comparisons = 2.

Time complexity :

<u>case</u>	<u>complexity</u>	<u>Reason</u>
Best case	$O(1)$	Target is first element.
Average case	$O(n)$	Target may appear anywhere.
Worst case	$O(n)$	Target is last or absent

For this problem:

Target position = 2

comparisons = 2

Run Time = $O(2) \Rightarrow O(n)$

Interpretation:

The target element appears immediately after the first element of the array. Linear Search requires only two comparisons with time complexity of $O(n)$

Thus, Linear search is suitable for small or unsorted dataset but is less efficient than Binary search for large dataset.